

BASESTACK ACADEMY · AWS CLOUD ACCELERATOR · COHORT 1

CAPSTONE PROJECT

Build Nigeria's Cloud Infrastructure

2 Weeks · 1,000 Points · 3 Scenarios · Full-Stack AWS

Overview

You have spent 8 weeks learning how to build on AWS. 40 daily labs. Every service hands-on. The capstone is not another lab — there is no walkthrough, no step-by-step. There is a business problem, a rubric, and 14 days. How you architect it is your engineering decision.

A Distinction submission proves you can read a requirement, design from first principles, deploy end-to-end, secure every layer, monitor it, document it, and justify every decision. That is the difference between a lab graduate and a cloud engineer.

Total Marks: 1,000 Points

This project is graded out of 1,000 points across 7 sections. The pass threshold is 600/1,000 (60%). Top 5 scores present live to the cohort on Day 21.

Every mark requires evidence. A claim without a screenshot, URL, or written justification scores zero.

Choose Your Scenario

Select ONE scenario. All three require the same AWS service layers and are graded on the same 1,000-point rubric. Only the business context differs.

SCENARIO A — OluPay 2.0

Lagos Fintech Scale-Up · Finance · High security · High throughput

OluPay has grown from 400,000 to 2 million users. Rebuild the entire infrastructure: multi-AZ VPC, EC2 web tier with Auto Scaling behind an ALB, S3 financial reports with lifecycle policies, Aurora MySQL for the payments ledger, RDS for KYC, DynamoDB merchant catalogue, ElastiCache Redis for OTP caching, and a fully serverless notification pipeline. Must handle salary day (25th of every month) without manual intervention and cost under \$200/month at steady state.

Key requirement: Every decision must be justified in both cost AND security terms.

SCENARIO B — EduTrack Nigeria

National Student Management Platform · Government-scale · Multi-tenant

Federal Ministry of Education contract: 500 universities, 3 million students, 120,000 staff. Handles registration, results, transcripts, and automated notifications. Must be multi-tenant (institution isolation at IAM and network layer), highly available, and accessible on low-bandwidth connections. Data sovereignty: all storage in af-south-1 with EU West DR.

Key requirement: Prove institution data isolation at the IAM and VPC layer in your submission.

SCENARIO C — LogiHaul

Pan-Nigeria Logistics Platform · Real-time · Unpredictable spikes

5,000 truck drivers across 36 states. Order placement, driver assignment, delivery tracking, billing, SMS/email alerts. Traffic spikes 50x during promos or flood events. Must scale elastically, survive AZ failure without data loss, confirm deliveries in <2 seconds. Budget: under \$150/month steady state, under \$400 at peak.

Key requirement: ASG config, DynamoDB capacity, and Lambda concurrency must all explicitly address the 50x spike scenario.

What You Must Build — All Scenarios

Layer	Services	Must Do
1 — Foundation & Identity	IAM, VPC	Custom VPC: 4 subnets (2 public, 2 private) across 2 AZs. IGW + route tables. Security Groups per tier. Bastion host. Minimum 3 IAM roles with least-privilege policies.
2 — Compute & Scaling	EC2, ALB, Auto Scaling	EC2 web tier behind ALB. ASG with documented scaling policy. Scale-out on CPU > 60%. Launch Template with UserData.
3 — Storage	S3	Static website (live URL). Versioning + lifecycle rule. SSE encryption. Cross-region replication. Pre-signed URL generated and tested.
4 — Databases	RDS/Aurora, DynamoDB, ElastiCache	One relational DB in private subnet (Multi-AZ or Read Replica justified). DynamoDB table with GSI + access pattern written before schema. ElastiCache Redis with TTL + (nil) expiry test.
5 — Serverless & Events	Lambda, API GW, SQS, SNS	Two Lambda functions, distinct responsibilities. HTTP API Gateway with live POST endpoint. SQS decoupling async process. SNS with confirmed email + alert triggered.
6 — Observability	CloudWatch	Dashboard (3+ services). Alarm with SNS action (not INSUFFICIENT_DATA). Log stream annotated. One Logs Insights query documented. Error simulation cycle.

Grading Rubric — 1,000 Points

Every criterion requires evidence: a screenshot, live URL, CLI output, or written justification.

"I built this" is not evidence.

SECTION 1 — FOUNDATION & IDENTITY (IAM + VPC)

150 pts

Criterion	Pts	Evidence Required	Pass Standard
Custom VPC — 4 subnets across 2 AZs	25	VPC console screenshot showing subnet list + AZ column	All 4 subnets visible, 2 different AZs confirmed
IGW attached + route tables correct	20	Route table screenshot — 0.0.0.0/0 → IGW on public only	Private subnets have NO IGW route
Security Groups: web, DB, bastion — correct inbound rules	25	Screenshot of each SG inbound rules	DB SG allows only from web SG on correct port. No 0.0.0.0/0 on port 22 except bastion.
Bastion SSH → private resource confirmed	20	Terminal: SSH to bastion then connection to private resource	Private instance IP visible in connected session
3+ IAM roles with least-privilege — custom policy shown	30	IAM roles list + one custom policy JSON	No AdministratorAccess. Each role scoped to its service only.
Architecture diagram — VPC layer complete	15	Excalidraw/draw.io showing VPC, subnets, SGs, IGW	All components present and connected correctly
AUTO-DEDUCTION: any DB/cache in public subnet	-25	N/A	Automatic — non-negotiable

SECTION 2 — COMPUTE & SCALING (EC2 + ALB + ASG)**150 pts**

Criterion	Pts	Evidence Required	Pass Standard
EC2 in private subnet + ALB serving HTTP 200	25	EC2 console + ALB DNS returning valid response in browser	ALB URL must load a response
ALB with healthy target group	25	ALB console + target group showing healthy targets	At least 1 healthy instance registered
ASG Min/Max/Desired documented + scaling policy	30	ASG console + scaling policy screenshot	Min ≥ 1 , Max ≥ 3 , policy trigger documented
Scale-out event in ASG Activity history	30	ASG Activity tab showing scale event	Log shows "Launching a new EC2 instance"
Launch Template with UserData serving web content	20	Launch template config + browser showing app via ALB URL	ALB DNS serves a page
Instance type choice justified in writing	20	Min 3 sentences in submission field	Mentions On-Demand vs Reserved tradeoff for scenario

SECTION 3 — STORAGE (S3)**100 pts**

Criterion	Pts	Evidence Required	Pass Standard
S3 static website live and publicly accessible	25	Live URL + screenshot of loaded page	Must return content — not Access Denied
Versioning enabled + lifecycle rule configured	20	Bucket Properties screenshot showing both enabled	Lifecycle rule must transition or expire at least one class
SSE-S3 or SSE-KMS encryption enabled	15	Bucket Properties > Default encryption screenshot	Encryption type visible and enabled
Cross-region replication to secondary region	20	Replication rule + destination bucket in different region	Rule shows Enabled status
Pre-signed URL generated + tested (expires ≤5 min)	20	CLI command used + URL tested before and after expiry	URL returns object. After expiry: Access Denied.

SECTION 4 — DATABASES (RDS/Aurora + DynamoDB + ElastiCache)

200 pts

Criterion	Pts	Evidence Required	Pass Standard
RDS/Aurora in private subnet — Public accessibility: NO	25	RDS Connectivity tab screenshot	Must show NO. Any YES scores 0 for this criterion.
RDS vs Aurora choice justified for scenario	20	Min 4 sentences — covers throughput, failover, cost, use case	All four dimensions addressed
Schema: 5+ columns + 3 sample rows inserted	20	SELECT * from bastion terminal	3 rows visible in CLI output
Multi-AZ or Read Replica — choice justified	15	Console showing standby/replica + written reason	Both valid if justified correctly
DynamoDB: access pattern WRITTEN before schema	25	Access pattern in submission field + table schema matches it	Pattern written first. Schema must serve the pattern.
DynamoDB GSI created and queried	20	GSI in Indexes tab + Query result using GSI	GSI must return items. Empty result = fail.
Query vs Scan both run — cost difference explained	15	Both screenshots + written explanation	Explanation correctly states Scan reads all items
ElastiCache Redis TTL pattern + (nil) expiry test	25	redis-cli SET/GET/TTL + (nil) expiry screenshot	(nil) must be visible. SET/GET only = 0 pts.
Cache-aside pattern explained step-by-step	15	Written in submission field	Must correctly name: hit, miss, write-through steps

SECTION 5 — SERVERLESS & EVENTS (Lambda + API GW + SQS + SNS)

250 pts

Criterion	Pts	Evidence Required	Pass Standard
Lambda 1 — primary business logic, active + no errors	30	Lambda console Active status + test returning 200	Executes without error on test invocation
Lambda 2 — separate responsibility from Lambda 1	25	Second Lambda console + different trigger/purpose described	Two distinct jobs documented. One function doing both = fail.
Each Lambda has least-privilege IAM role (no * resources)	25	IAM role JSON for each Lambda	Specific resource ARNs. No wildcard "*".
API Gateway HTTP API — live POST endpoint working	30	curl/Postman showing HTTP 200 + JSON body	Must be HTTP API not REST API.
API → Lambda → DynamoDB write confirmed end-to-end	30	DynamoDB Explore items showing record from API call	Record appears after API call — not pre-inserted.
SQS decoupling Lambda 1 from Lambda 2	30	SQS queue console + Lambda 2 SQS trigger configured	Trigger visible on Lambda 2. Batch size documented.
SQS Visibility Timeout set correctly + justified	20	Queue settings screenshot + written explanation of value	Value ≥ expected Lambda duration. Explanation required.
SNS confirmed email subscription + alert received	30	Subscription showing Confirmed + email in inbox	Email must actually arrive. Pending = 0 pts.
Cost estimate: serverless layer at scenario scale	30	Table: Lambda + API GW + SQS + SNS costs at scenario users	Must use scenario user counts, not defaults.

SECTION 6 — OBSERVABILITY (CloudWatch)**100 pts**

Criterion	Pts	Evidence Required	Pass Standard
Dashboard with metrics from 3+ services	25	Dashboard screenshot with named widgets from different services	Lambda + API Gateway + one DB metric minimum
Alarm with SNS action — in OK or ALARM state	25	Alarm console showing threshold + SNS action	NOT in INSUFFICIENT_DATA at submission time
Lambda log stream reviewed + annotated	20	Log stream screenshot with student annotation of key lines	START/END/REPORT lines identified by student
Logs Insights query run + result documented	20	Query text + result screenshot	Query must filter or aggregate — not just list all
Error simulation: force error → find in logs → fix	10	Error screenshot + log showing it + corrected code	Full cycle: introduced → found → fixed

SECTION 7 — ARCHITECTURE & DOCUMENTATION**50 pts**

Criterion	Pts	Evidence Required	Pass Standard
Complete architecture diagram — all layers labelled	20	PNG exported in submission	All 6 layers present. Data flow arrows correct.
GitHub repo with README	15	Repo URL + README screenshot	README: overview, diagram, services, how to test
Portfolio post published — LinkedIn or Twitter	15	Screenshot of published post (public)	Post includes diagram or screenshot

Scoring Summary

Section	Points	Minimum to Pass
1 — Foundation & Identity (IAM + VPC)	150	90 pts (60%)
2 — Compute & Scaling (EC2 + ALB + ASG)	150	90 pts (60%)
3 — Storage (S3)	100	60 pts (60%)
4 — Databases (RDS/Aurora + DynamoDB + Redis)	200	120 pts (60%)
5 — Serverless & Events	250	150 pts (60%)
6 — Observability (CloudWatch)	100	60 pts (60%)
7 — Architecture & Documentation	50	30 pts (60%)
TOTAL	1,000	600 pts — Pass Threshold

Score Range	Grade	Classification
900 – 1,000	A+	Distinction — Production-Ready Engineer
800 – 899	A	Distinction — Cloud Engineer
700 – 799	B	Merit — Junior Cloud Engineer
600 – 699	C	Pass — Cloud Practitioner Level
Below 600	F	Fail — Resubmission within 7 days

Automatic Deductions (max -90 pts total)

- 25 pts: Any database or cache placed in a public subnet
- 25 pts: Any IAM role with AdministratorAccess or wildcard (*) resource
- 15 pts: Default VPC used instead of custom VPC
- 15 pts: Instance-specific Aurora endpoints used instead of cluster endpoints
- 10 pts: S3 data bucket with public read enabled (portfolio bucket excepted)

Submission Rules

- Submit via BaseStack LMS only — not WhatsApp, email, or DM
- Screenshot naming convention: S1-C1.png (Section 1, Criterion 1), S5-C4.png, etc.
- Live URLs paste directly into the form field — do not screenshot URLs
- Written justifications go in form text fields — not in a separate document
- Do NOT delete any AWS resources before grading is confirmed

Cost Warning — Delete These Promptly After Screenshots

Aurora db.t3.medium: ~\$2.50/day — delete immediately after grading confirmed

ElastiCache cache.t3.micro: ~\$0.40/day — delete immediately after grading confirmed

EC2 (if not Free Tier eligible): ~\$0.50–2.00/day — terminate after grading confirmed

Everything else (Lambda, SQS, SNS, DynamoDB On-Demand, S3): negligible or free tier